Temporary Material Configurations

Unbuilding Vancouver's houses for the missing middle

Ruijie Liu · Speaking script with slide visuals

32	1,565	10:55
spoken slides	spoken words	planned, limit 11:00

Outline

1	The question	slides 1 to 3	0:41
2	Why now	slides 4 to 10	1:29
3	What others know	slides 11 to 14	1:30
4	How I work	slides 15 to 19	2:03
5	First results	slides 20 to 23	1:22
6	The proposal	slides 24 to 29	2:31
7	The plan	slides 30 to 32	1:19

How to read this script

Each slide has its own page. The spoken paragraph comes first, with every sentence numbered: each number is one click, and the slide lights the part that sentence names. The slide as the audience sees it follows, with its on-screen message and, at the foot of the page, the small source note shown on the slide. Where the slide cites a book, paper or project, images from the original source sit under the visual. Times assume about 147 words a minute. Small yellow tags are delivery cues, not spoken: *pause* is a short beat, *slower* slows the next words, *keep going* means the sentence runs on past the click. An underlined word takes the stress.

1 Title

1 Good afternoon. I'm Ruijie, and my project is **Temporary Material Configurations**: pause what architecture can do with houses Vancouver takes down for denser housing.



On screen: Temporary Material Configurations. Unbuilding Vancouver's houses for the missing middle.

2 Topic statement

1 My topic statement: pause I investigate how architectural design might change when buildings are understood as temporary material configurations, whose components can be evaluated, salvaged and reassembled during Vancouver's transition toward denser housing. pause Temporary describes the **arrangement**; materials can outlast it.



On screen: I investigate how architectural design might change when buildings are understood as temporary material configurations whose components can be evaluated, salvaged and reassembled during Vancouver's transition toward denser housing.

Source note on the slide: Topic statement

3 A permanent object

1 We build most houses as **permanent** objects: glued, sealed and poured together, so at the end pause the only way out is the excavator.

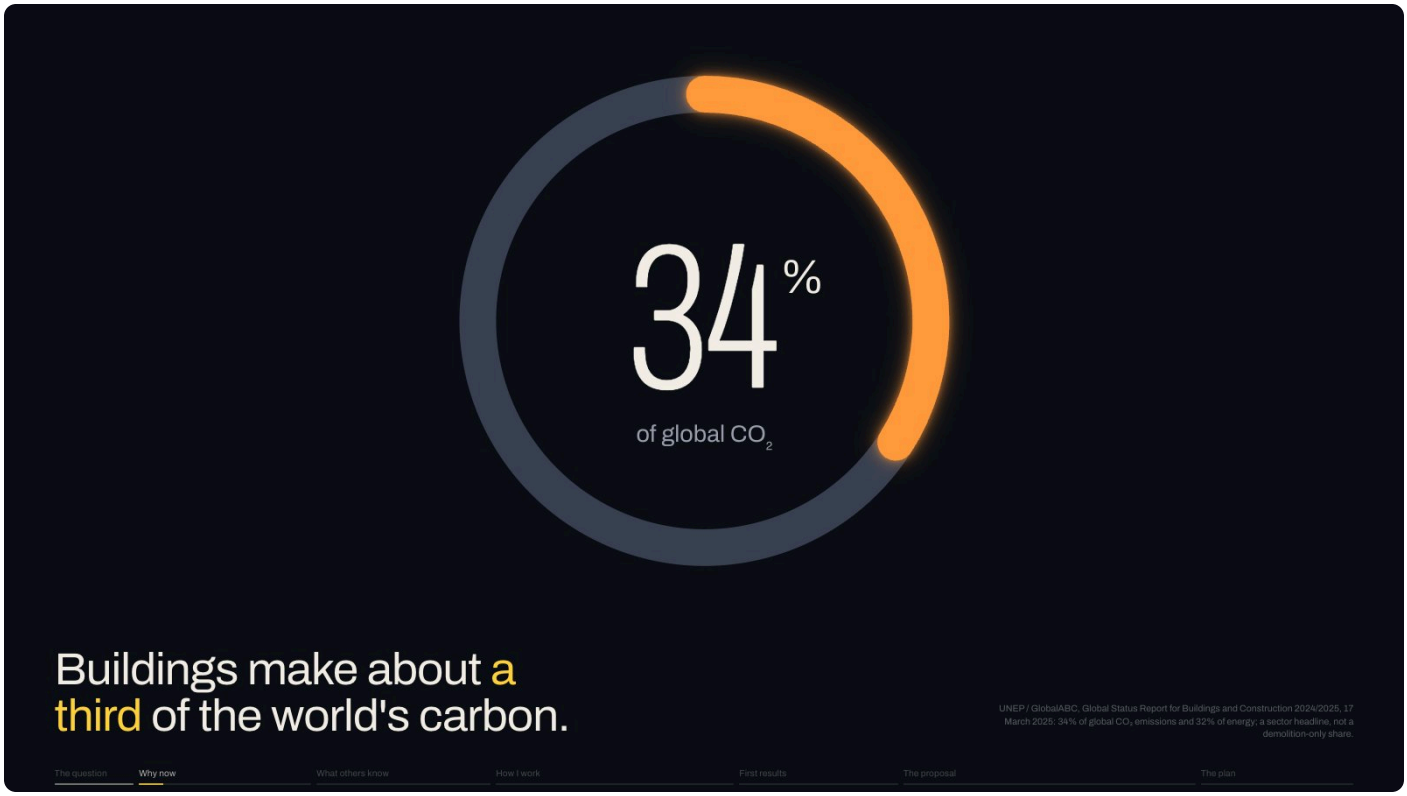


On screen: A permanent object can only be destroyed.

Source note on the slide: Drawing: R. Liu with Astra 6. Excavator after "Excavator" by RhymOne (Sketchfab, CC BY 4.0) and lorry after "Cargo Truck" by J-Toastie (Poly Pizza, CC BY 3.0), rendered in the board camera and adapted; workers after Quaternius (CC0).

4 Global carbon

1 Why now? pause I'd start with carbon: the UN Environment Programme finds that buildings and construction make about **a third** of the world's emissions.



On screen: Buildings make about a third of the world's carbon.

Source note on the slide: UNEP / GlobalABC, Global Status Report for Buildings and Construction 2024/2025, 17 March 2025: 34% of global CO_2 emissions and 32% of energy; a sector headline, not a demolition-only share.

5 Landfill

1. Closer to home, building and demolition material makes up about **a third** of what Metro Vancouver landfills.



On screen: And about a third of what our region landfills.

Source note on the slide: Metro Vancouver, Construction and Demolition Waste: construction, renovation and demolition material is about a third of what goes to its landfill; 2022 waste composition study: wood and plastic are its largest parts. Drawing: R. Liu with Astra 6.

6 A home every three hours

1 And homes keep coming down: pause across the region, about one every **three hours**.

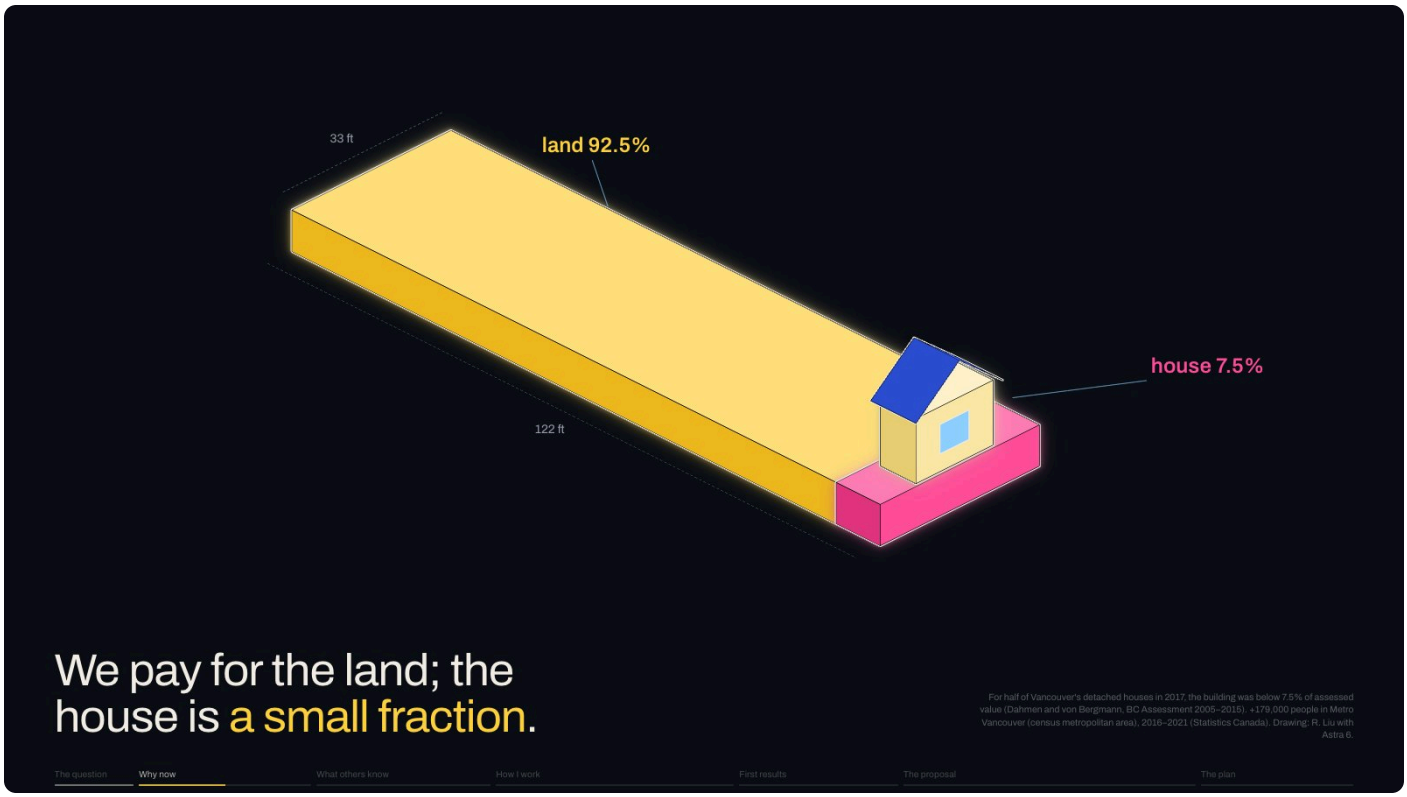


On screen: A home comes down every three hours.

Source note on the slide: Metro Vancouver average, ground-oriented dwellings, 2012–2022: 2,621 a year (Light House, A Blueprint for Change, 2023). 8,760 h ÷ 2,621 = 3.3 h.

7 We pay for the land

1. Houses come down because we mostly pay for **land**.
2. Joseph Dahmen and data scientist Jens von Bergmann found that for half the city's detached houses, the building is **a small fraction** of its value.



On screen: We pay for the land; the house is a small fraction.

Source note on the slide: For half of Vancouver's detached houses in 2017, the building was below 7.5% of assessed value (Dahmen and von Bergmann, BC Assessment 2005–2015). +179,000 people in Metro Vancouver (census metropolitan area), 2016–2021 (Statistics Canada). Drawing: R. Liu with Astra 6.

8 Why now: R1-1 zoning

1. Zoning explains the timing.
2. Until 2023, most lots allowed a house, suite and laneway home.
3. Since 2023, new zoning allows multiplexes: pause six to eight homes a lot.
4. A multiplex must be new, so the old house comes down first. pause The question is what happens to its **parts**.



On screen: Since 2023, one lot can hold six to eight homes, after a teardown.

Source note on the slide: City of Vancouver: R1-1 replaced nine RS zones, 17 Oct 2023; multiplex up to 1.0 FSR, 3–6 strata or 8 secured rental units (557 m² sites for 6–8). Before: house, suite and laneway home, 0.7 FSR. A multiplex is new construction (Low Density Housing Options guide, Dec 2025, p. 18). About 370 multiplex applications accepted by March 2025 (City memo, 29 May 2025). Drawing: R. Liu with Astra 6.

9 Who it is for

1 slower The new homes matter to people like me: newcomers who share a house with several others to afford the city.



On screen: I share a house with several others to afford this city.

Source note on the slide: The quote is my own. A purpose-built two-bedroom rents for $2,363 a month on average (CMHC Rental Market Report, Dec 2025).

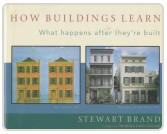
10 Why homes

1. I chose homes, though shops and factories come down too. pause
2. In How Buildings Learn, Stewart Brand sorts buildings by how they change.
3. Commercial buildings change **fast**; keep going
4. institutions **resist** change; keep going
5. homes change **steadily**, keeping a record of the family inside.
6. So homes are my test case, and that is deliberate.



On screen: Why homes? They are the steadiest changers.

From the sources

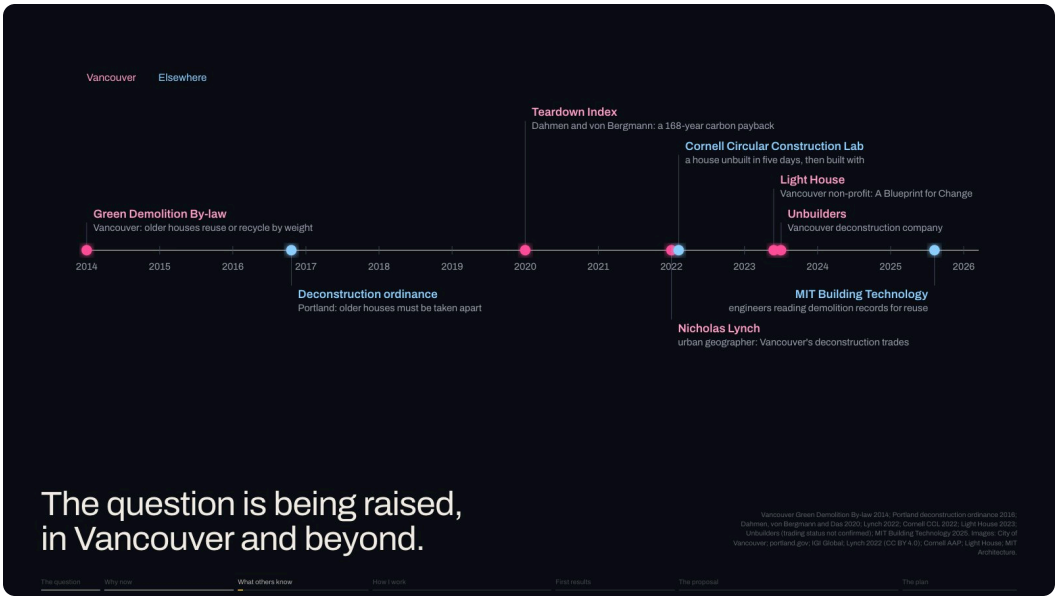


How Buildings Learn
Stewart Brand, Viking
1994, cover

Source note on the slide: Stewart Brand, How Buildings Learn, 1994, ch. 1, "Flow", after Sim Van der Ryn.

11 Who is asking

1. I'm not the first to ask. pause
2. Vancouver and Portland regulate demolition waste.
3. Dahmen and von Bergmann weighed the carbon of replacement.
4. Geographer Nicholas Lynch studied Vancouver's deconstruction trades, and Cornell's Circular Construction Lab rebuilt with a house it took apart.
5. The non-profit Light House and the deconstruction firm Unbuilders pushed for salvage.
6. MIT engineers read demolition records for reusable structure.



On screen: The question is being raised, in Vancouver and beyond.

From the sources



Green Demolition By-law No. 11023
City of Vancouver
2014, cover, consolidated 2025



Deconstruction requirements
City of Portland
2016, portland.gov



Examining the Environmental Impacts of Materials and Buildings
Dahmen, von Bergmann and Das, Teardown Index chapter
2020, IGI Global, cover



Unbuilding the city
Nicholas Lynch, Environment and Planning A
2022, open access, CC BY 4.0



Catherine Commons, taken apart
Cornell Circular Construction Lab
2022, Cornell AAP



A Blueprint for Change
Light House
2023, report cover



Dunbar house, taken apart by Unbuilders
City of Vancouver case study
2019, City of Vancouver

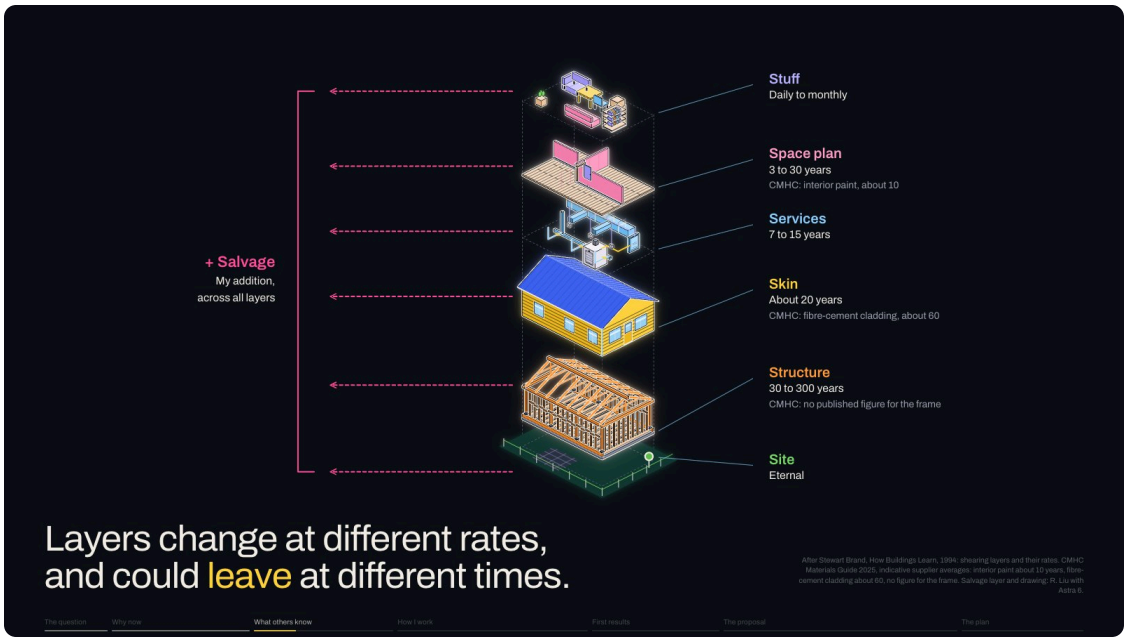


Build Reuse conference
MIT Building Technology
2025, MIT Architecture

Source note on the slide: Vancouver Green Demolition By-law 2014; Portland deconstruction ordinance 2016; Dahmen, von Bergmann and Das 2020; Lynch 2022; Cornell CCL 2022; Light House 2023; Unbuilders (trading status not confirmed); MIT Building Technology 2025. Images: City of Vancouver; portland.gov; IGI Global; Lynch 2022 (CC BY 4.0); Cornell AAP; Light House; MIT Architecture.

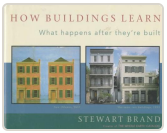
12 Layers change at different rates

1. Brand's six layers change at different rates, from site to stuff.
2. CMHC, Canada's housing agency, gives some of them lifespans: finishes last years, keep going
3. cladding, decades, keep going
4. the structural frame, pause **no figure at all**.
5. To his six I add **salvage**: pause when a configuration ends, each layer could leave on its own clock.



On screen: Layers change at different rates, and could leave at different times.

From the sources



How Buildings Learn
Stewart Brand, Viking
1994, cover

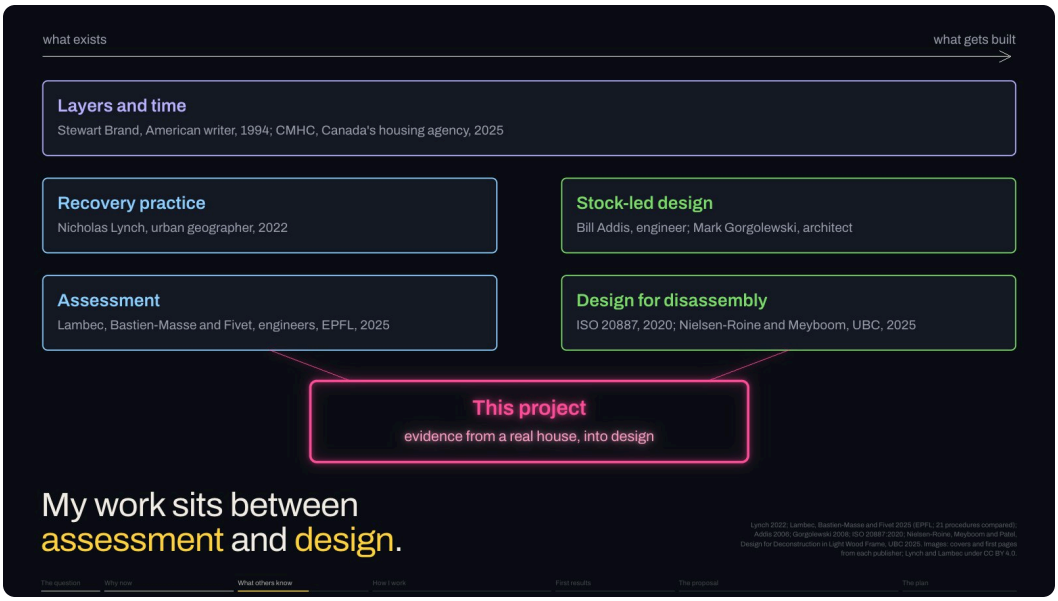


Housing Design Catalogue: Materials Guide
CMHC with Ha/f Climate Design
2025, first page

Source note on the slide: After Stewart Brand, How Buildings Learn, 1994: shearing layers and their rates. CMHC Materials Guide 2025, indicative supplier averages: interior paint about 10 years, fibre-cement cladding about 60, no figure for the frame. Salvage layer and drawing: R. Liu with Astra 6.

13 Position

1. Four kinds of work shape my position.
2. Recovery practice shows salvage depends on skills, timing and markets.
3. Engineers at EPFL, the Swiss Federal Institute of Technology in Lausanne, compare ways of judging reuse.
4. Stock-led design, from engineer Bill Addis and architect Mark Gorgolewski, designs from what exists.
5. Design for disassembly, now standardised, plans the next taking apart; at SALA, AnnaLisa Meyboom and Kaia Nielsen-Roine adapted it to light wood frame.
6. My work sits **between** assessment and design.



On screen: My work sits between assessment and design.

From the sources



Unbuilding the city
Nicholas Lynch, Environment and Planning A
2022, open access, CC BY 4.0



Procedures used to determine the reuse potential of building elements
Lambec, Bastien-Masse and Fivet, EPFL
2025, Frontiers, open access, CC BY 4.0



Building with Reclaimed Components and Materials
Bill Addis, Earthscan
2006, cover



Designing with reused building components
Mark Gorgolewski
2008, Building Research and Information



ISO 20887, Design for disassembly and adaptability
ISO
2020, iso.org

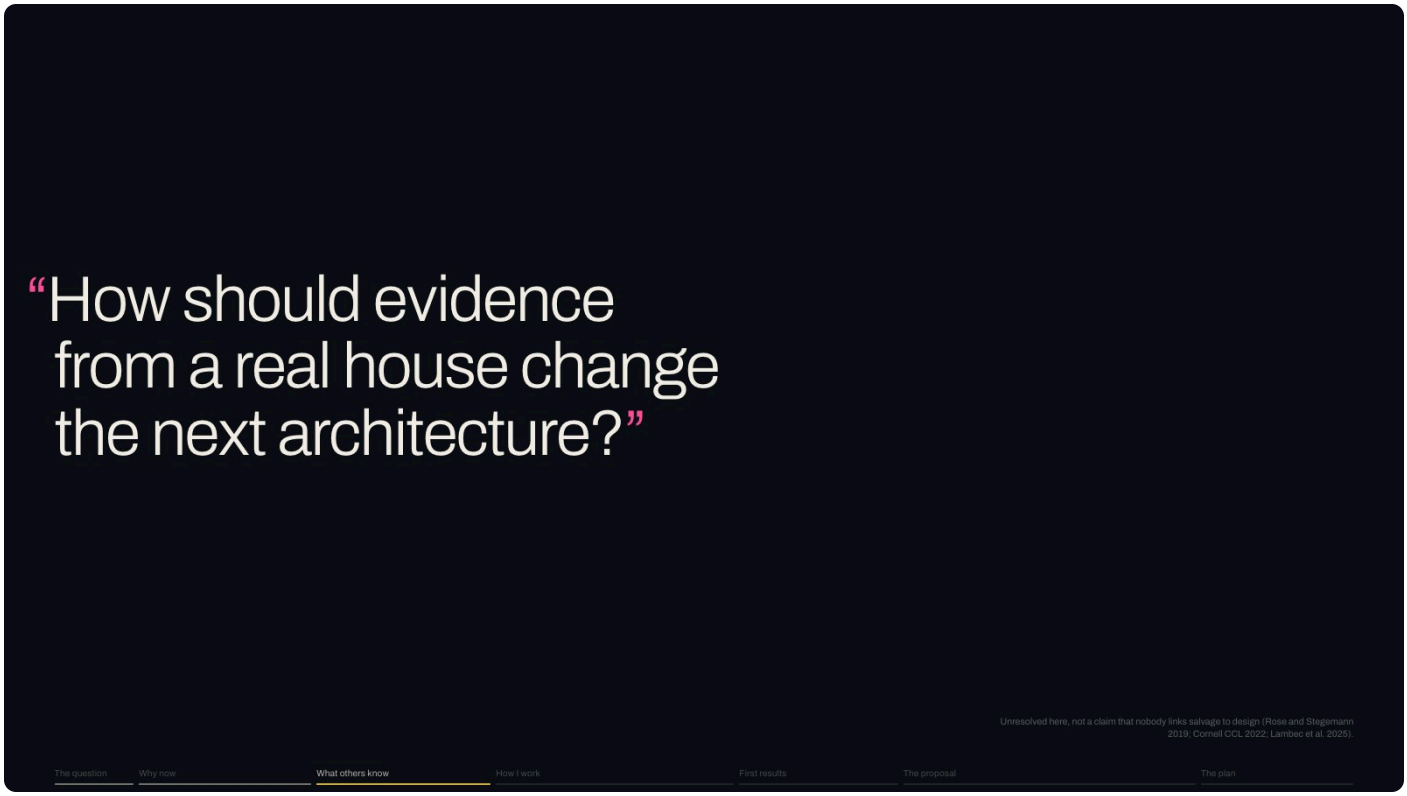


Design for Deconstruction in Light Wood Frame
Nielsen-Roine, Meyboom and Patel, UBC
2025, guidebook cover

Source note on the slide: Lynch 2022; Lambec, Bastien-Masse and Fivet 2025 (EPFL; 21 procedures compared); Addis 2006; Gorgolewski 2008; ISO 20887:2020; Nielsen-Roine, Meyboom and Patel, Design for Deconstruction in Light Wood Frame, UBC 2025. Images: covers and first pages from each publisher; Lynch and Lambec under CC BY 4.0.

14 The gap

1 So here's the gap. pause We can assess parts, and design buildings that come apart. pause But we **don’t yet know** how evidence from a real Vancouver house should change the next architecture, or where parts wait in between.

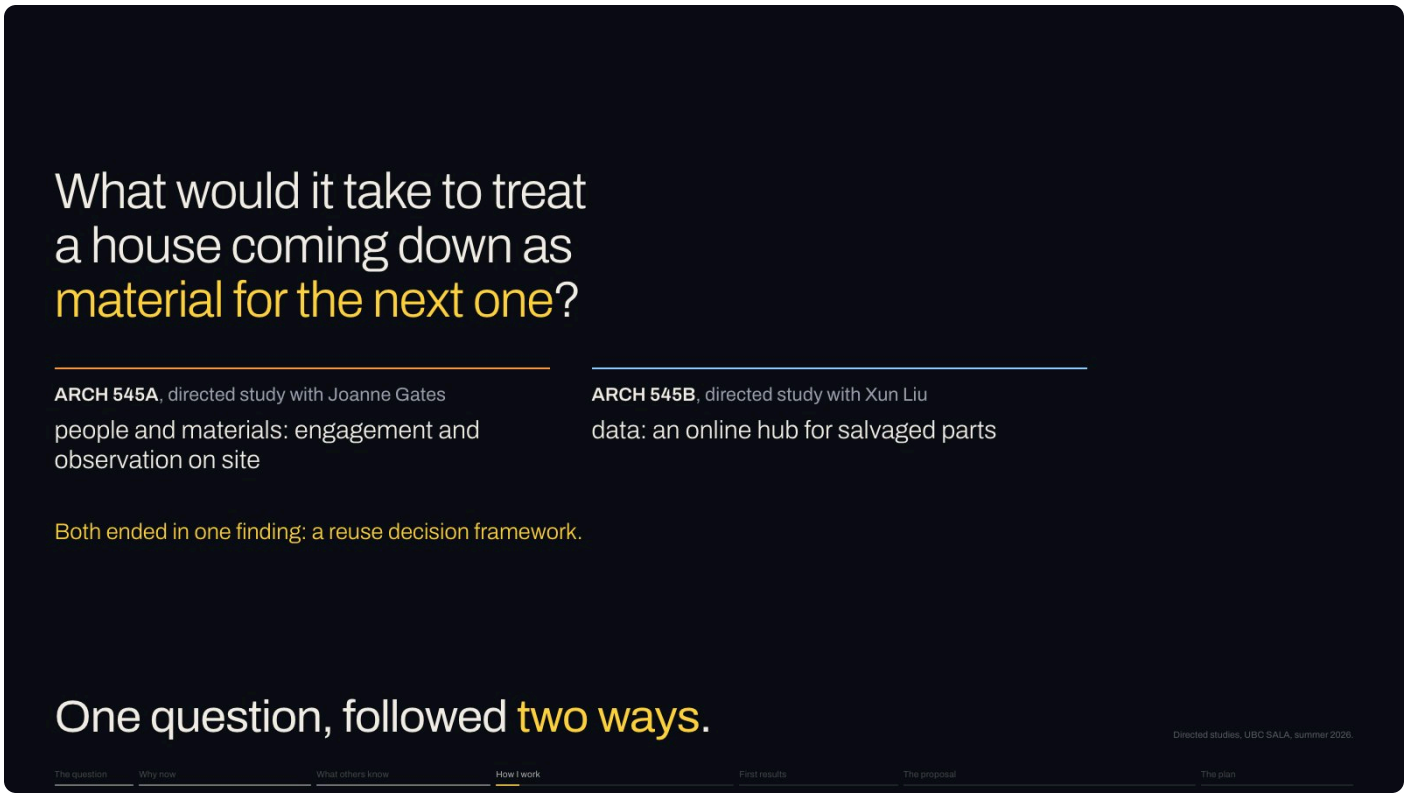


On screen: How should evidence from a real house change the next architecture?

Source note on the slide: Unresolved here, not a claim that nobody links salvage to design (Rose and Stegemann 2019; Cornell CCL 2022; Lambec et al. 2025).

15 Where it began

1. I came to this gap by watching sound houses come down as waste. pause Could they become the **next building**?
2. This summer I followed that through two directed studies.

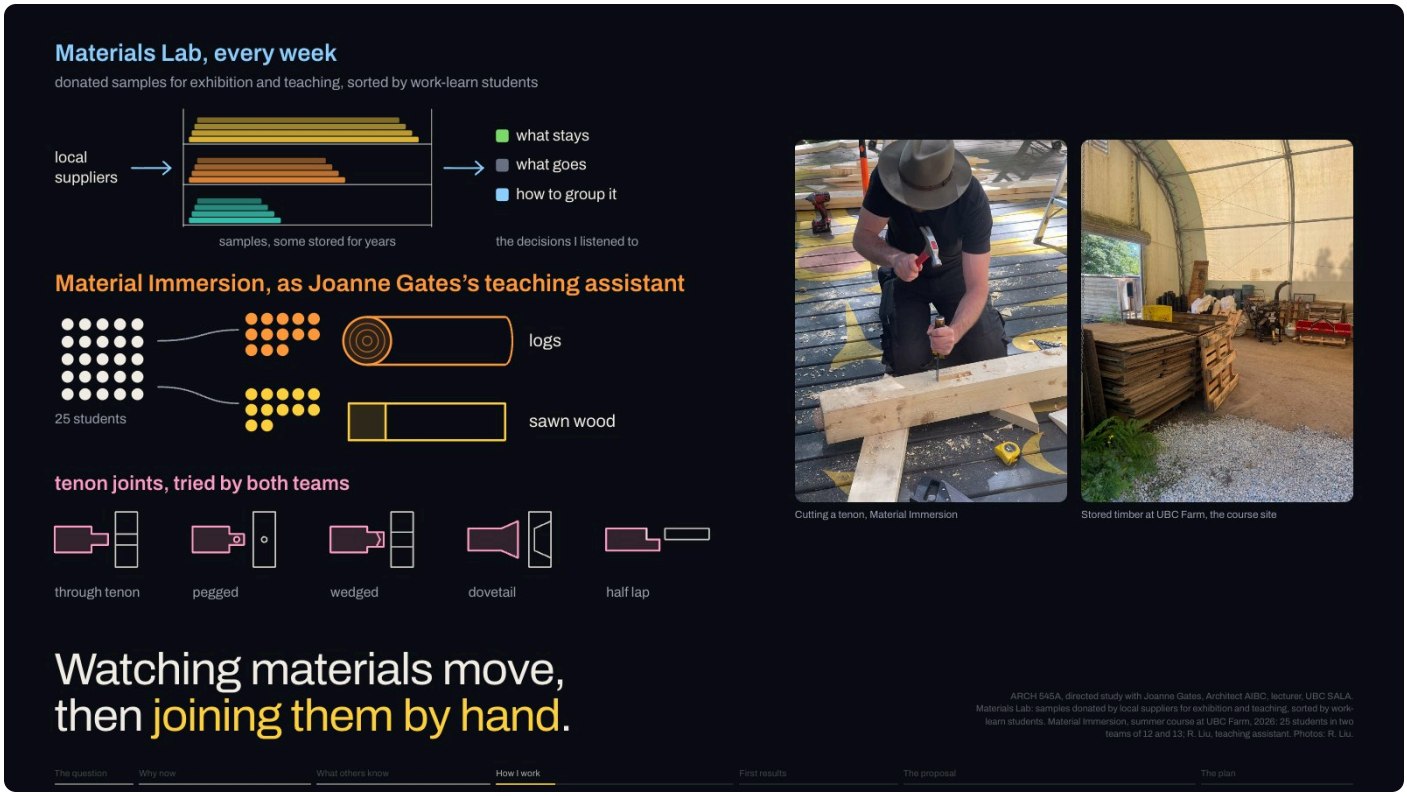


On screen: One question, followed two ways.

Source note on the slide: Directed studies, UBC SALA, summer 2026.

16 Study one: on site

1. The first, with Joanne Gates, was observation on site.
2. Weekly at the Materials Lab, I listened as work-learn students sorted donated samples: pause what stays, what goes, how to group them.
3. As her teaching assistant on Material Immersion, I helped oversee twenty-five students, one team with logs, one with sawn wood, keep going
4. both trying tenon joints, where my interest in joints began.



On screen: Watching materials move, then joining them by hand.

Source note on the slide: ARCH 545A, directed study with Joanne Gates, Architect AIBC, lecturer, UBC SALA. Materials Lab: samples donated by local suppliers for exhibition and teaching, sorted by work-learn students. Material Immersion, summer course at UBC Farm, 2026: 25 students in two teams of 12 and 13; R. Liu, teaching assistant. Photos: R. Liu.

17 The pagoda study

1. Joints then took me to Shanxi, in China, to timber buildings the architects Liang Sicheng and Lin Huiyin drew.
2. In the Yingxian Pagoda, a timber tower from 1056, bracket sets called dougong lock short pieces without nails or glue, pause yet carry the roof far out.
3. A short piece isn't waste **if a joint can make it work**. pause hold up the printed set Please try it.

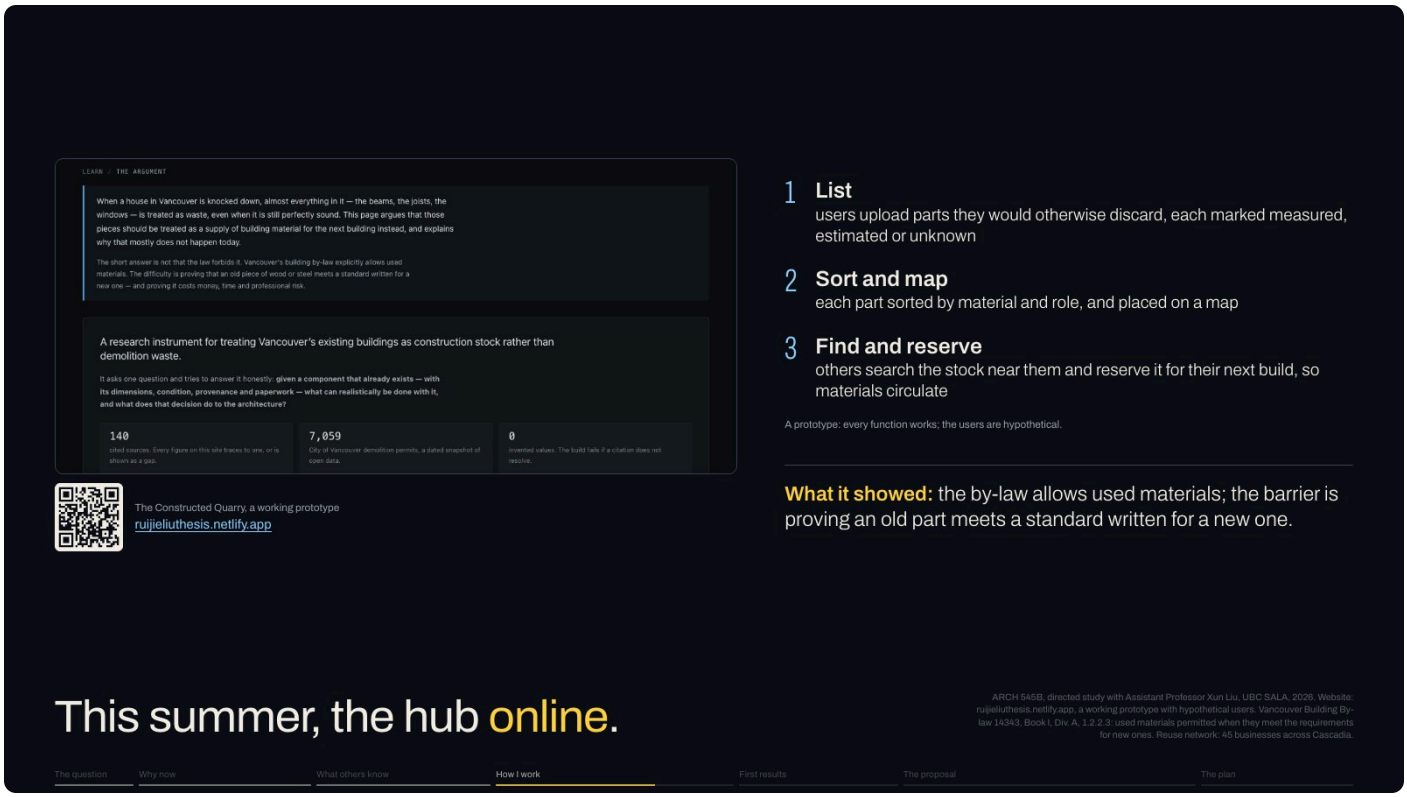


On screen: Short pieces, joined, carry the roof, and come apart again.

Source note on the slide: UNESCO Tentative List submission, 2013: built 1056, about 67 m high, over 15,400 timber elements. Liang Ssu-ch'eng (Liang Sicheng), A Pictorial History of Chinese Architecture, ed. W. Fairbank, 1984; Liang and Lin Huiyin's Shanxi surveys for the Society for Research in Chinese Architecture, 1930s. Bracket set on screen: original, unmeasured teaching model by R. Liu with Astra 6, after the Yingzao Fashi (1103) and two Bilibili assembly studies (BV1Kg1TYJEDd, BV1WX4y127Pq). Printed model: 宋式八铺作斗拱 by user_3800505495, MakerWorld, Standard Digital File License, personal printing.

18 Study two: an online hub

1. The second, with Xun Liu, used AI to gather, structure and show data.
2. I built a prototype online hub: pause users list parts they'd discard, sorted and mapped, and others reserve them for a next build, so materials circulate. Its users are **hypothetical**.
3. It showed the barrier: pause the by-law allows used materials, but **proving** an old part meets a new standard costs time and money.

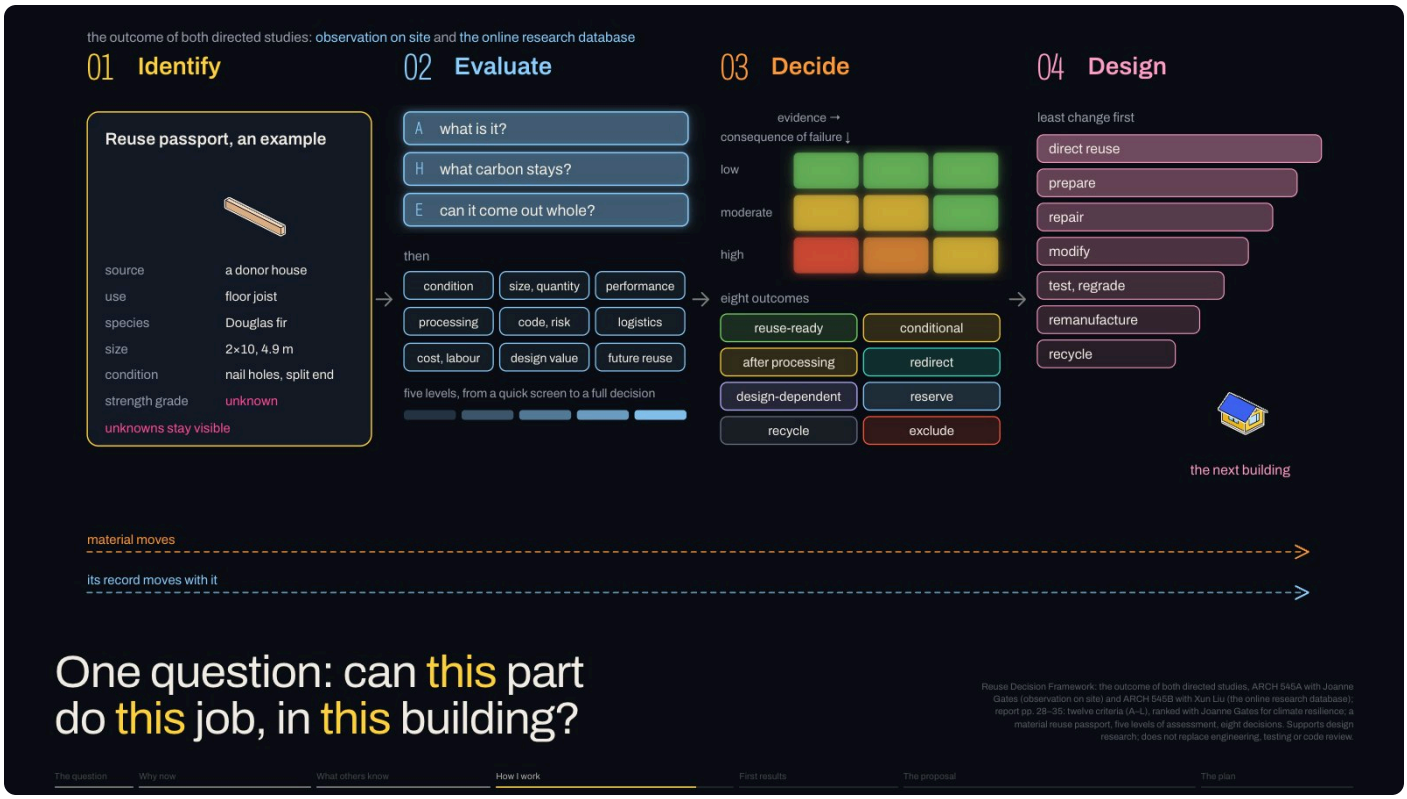


On screen: This summer, the hub online.

Source note on the slide: ARCH 545B, directed study with Assistant Professor Xun Liu, UBC SALA, 2026. Website: ruijieliuthesis.netlify.app, a working prototype with hypothetical users. Vancouver Building By-law 14343, Book I, Div. A, 1.2.2.3: used materials permitted when they meet the requirements for new ones. Reuse network: 45 businesses across Cascadia.

19 How I will decide

1. Both studies led to one outcome. pause From observation on site and the online research database, I built the reuse decision framework. It asks: pause can **this** part do **this** job, **here**?
2. First, identify: a passport of what we know about each part, and what we don't.
3. Then evaluate it for that use: what it is, its carbon, whether it comes out whole.
4. Then decide, weighing evidence against the consequence of failure.
5. And design, least change first, a path into the next building, its record travelling with it.

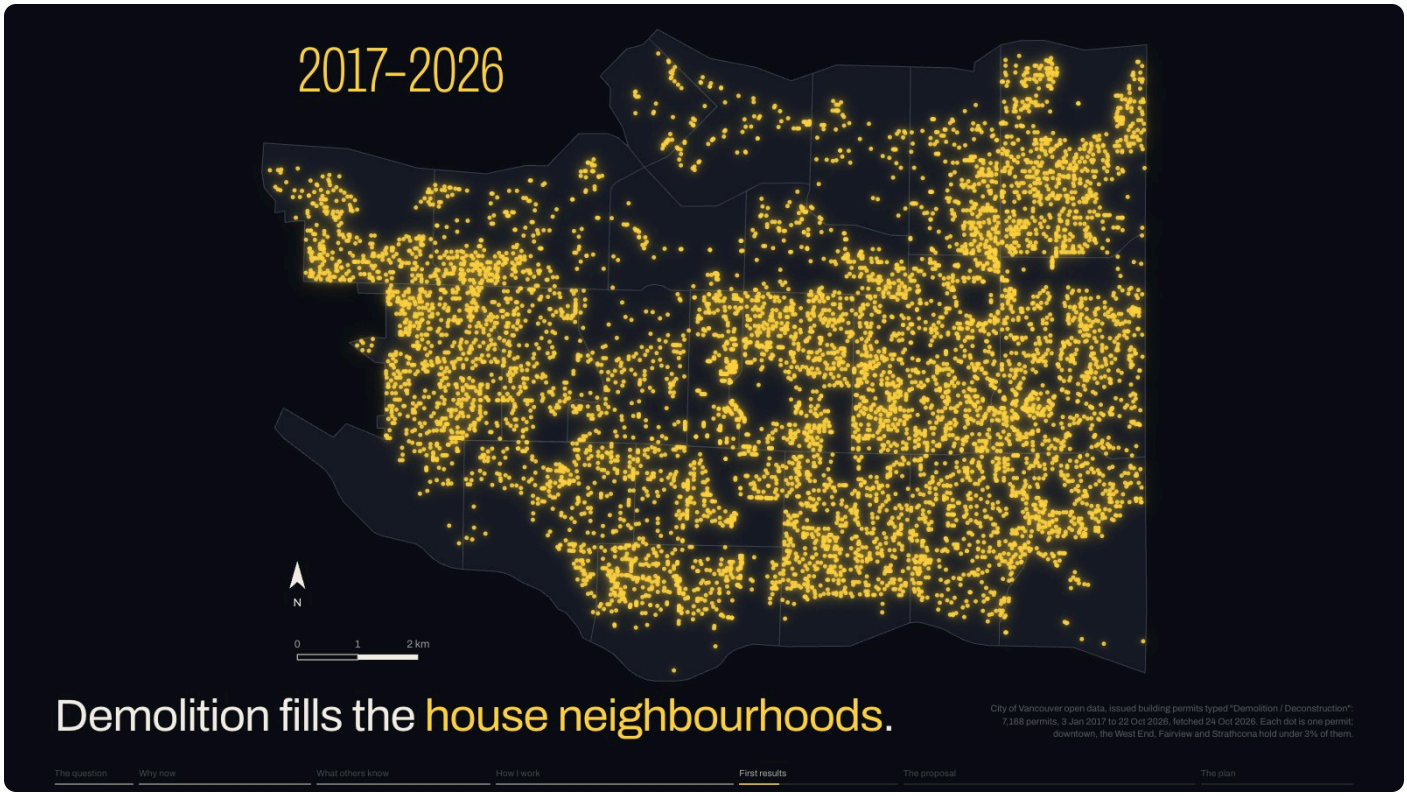


On screen: One question: can this part do this job, in this building?

Source note on the slide: Reuse Decision Framework: the outcome of both directed studies, ARCH 545A with Joanne Gates (observation on site) and ARCH 545B with Xun Liu (the online research database); report pp. 28–35: twelve criteria (A–L), ranked with Joanne Gates for climate resilience; a material reuse passport, five levels of assessment, eight decisions. Supports design research; does not replace engineering, testing or code review.

20 Every demolition permit

1 My first results map every demolition permit since 2017. pause They fill the house neighbourhoods, south and east, and almost **skip downtown**.



On screen: Demolition fills the house neighbourhoods.

Source note on the slide: City of Vancouver open data, issued building permits typed "Demolition / Deconstruction": 7,168 permits, 3 Jan 2017 to 22 Oct 2026, fetched 24 Oct 2026. Each dot is one permit; downtown, the West End, Fairview and Strathcona hold under 3% of them.

21 No requirement

1. Then I read what permits ask.
2. Over half ask **nothing**, because the Green Demolition By-law targets only older houses.
3. The rest name a share to reuse or recycle, by weight, keep going
4. the highest for character houses.



On screen: Most permits ask for little, or nothing.

Source note on the slide: Tier parsed from permit text, so counts are a floor: 3,831 none stated (53%), 2,651 at 75% (pre-1950, 37%), 686 at 90% (character houses, 10%). Green Demolition By-law No. 11023. Chart: R. Liu with Astra 6.

22 Recycled is not reused

1. By weight, reuse and recycling count **the same**.
2. Portland, which requires deconstruction, finds reuse a small share of what leaves the site.
3. An Oregon study of thirty-six homes **confirms it**: about a quarter was salvaged.
4. Vancouver counts the two together, so the city publishes no reuse rate. pause That **empty bar** is where my project starts.



On screen: Recycled by weight is not reused.

Source note on the slide: City of Portland BPS, Deconstruction Program, Sept 2025 (per project, about 4 t reused and 19 t to recovery; foundations excluded); Oregon DEQ 2019, 36 homes, about a quarter salvaged. Vancouver's by-law counts reuse and recycling together by weight. Chart: R. Liu with Astra 6.

23 Why wood first

1. To see where reuse should start, I looked at one best case the City documented: pause a 1930 Dunbar house, taken apart by hand.
2. The concrete was heaviest and was crushed for road fill; flooring, doors and trim found buyers.
3. But of nineteen tonnes of lumber, **three quarters** were burned for energy. The City notes pieces under four feet can only be recycled.
4. Across the region, wood is about half of building waste thrown away.
5. The hub takes the whole house; pause my research starts with its **wood**, and joints for short pieces.



On screen: Wood first: even by hand, most of the lumber was burned.

From the sources

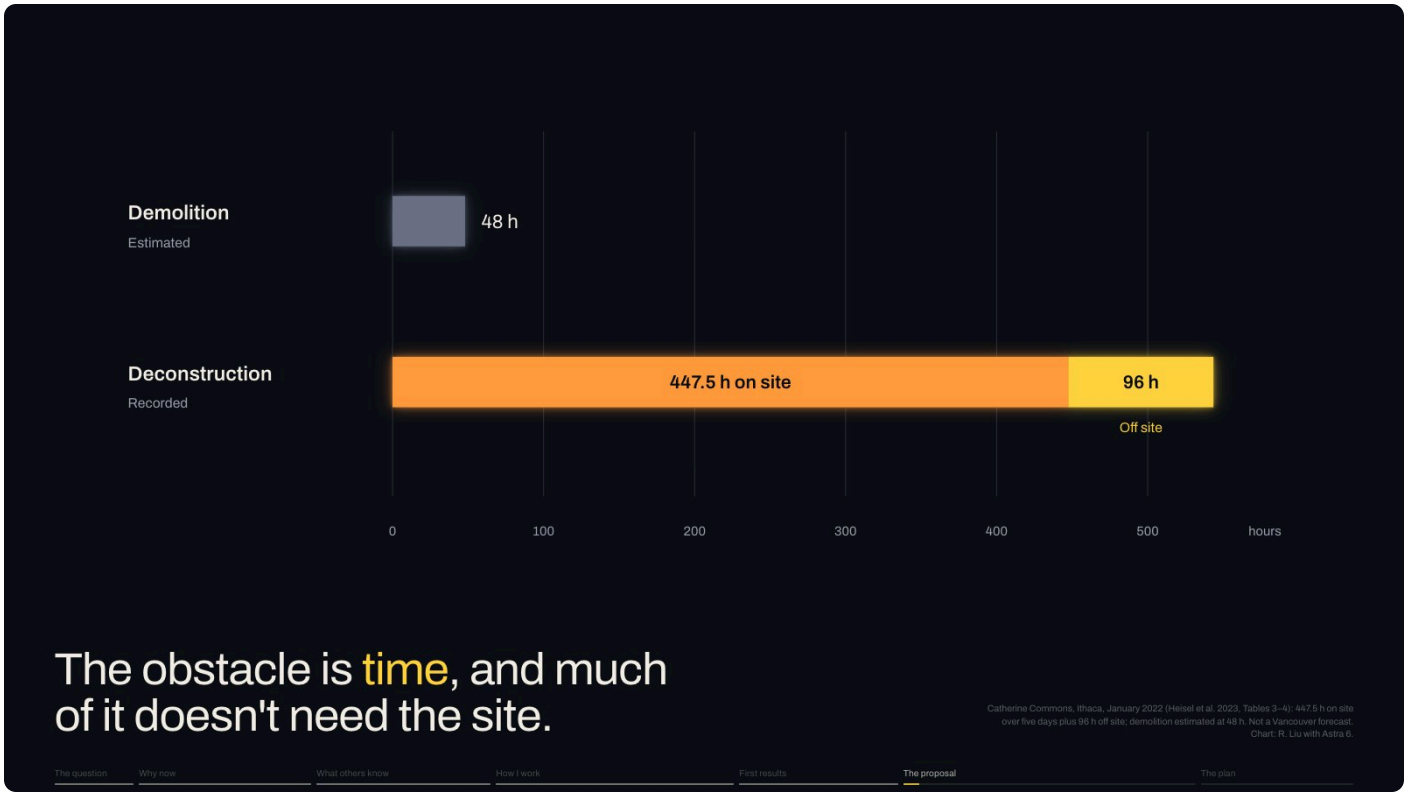


Dunbar house, taken apart by hand
City of Vancouver case study
2019, City of Vancouver

Source note on the slide: City of Vancouver, Deconstruction case study: Dunbar (a 1930 house, taken apart in 17 days in 2019): concrete 39,520 kg, crushed for road fill; lumber 19,220 kg (framing and boards; most pre-1950 homes are old growth, per the case study and Teshnizi 2015), 25% reused, 74% to waste-to-energy, 1% landfill ("lumber shorter than 4' can only be recycled"); 21% reused and 77% recycled overall, so burned lumber counted as recycled; flooring 5,549 kg, 85% reused; plaster 3,090 kg, recycled; fixtures reused through Habitat for Humanity. Metro Vancouver, 2022 Construction and Demolition Waste Composition Study (Dillon, 2023): wood of all kinds is 48% by weight of the 372,000 t disposed of in 2021. Joint: schematic, dowel-laminated, no glue. Chart: R. Liu. More on print 3, Why wood first.

24 The obstacle is time

1. Short pieces are one reason so little is reused. Cornell's lab points to another: pause **time**.
2. Taking it apart took about ten times the labour of demolition; on Vancouver land, each day costs money.
3. But much of that slow work needs a roof and bench, **not the site**.



On screen: The obstacle is time, and much of it doesn't need the site.

Source note on the slide: Catherine Commons, Ithaca, January 2022 (Heisel et al. 2023, Tables 3–4): 447.5 h on site over five days plus 96 h off site; demolition estimated at 48 h. Not a Vancouver forecast. Chart: R. Liu with Astra 6.

25 What exists, and what is missing

1. A place like this exists, pause but only in pieces.
2. Around Vancouver, firms take houses apart and stores resell parts, loosely organised, Lynch found.
3. Depots from Portland to Basel resell parts; in Winterthur, Switzerland, an extension was built largely from reused parts.
4. Here, salvaged wood has no grading rule, so only engineers can approve it for load.
5. The City backed a hub in 2018; the closest thing today, Habitat for Humanity's Rebuild Hub, gathers salvage but sells **outside the city**.



On screen: Others do single steps; none does them in one public place.

From the sources



Rebuild Hub salvage
Habitat for Humanity Greater Vancouver
2020, habitatgv.ca



Dunbar house, taken apart
City of Vancouver case study
2019, City of Vancouver



ReBuilding Center
Portland
rebuildingcenter.org



Salvage Lumber Warehouse, opening
Seattle Public Utilities
2026, seattle.gov



Rotor DC
Brussels
2022, photo Pascal Broze, rotordb.org



Bauteilbörse Basel
Basel
bauteilboerse-basel.ch

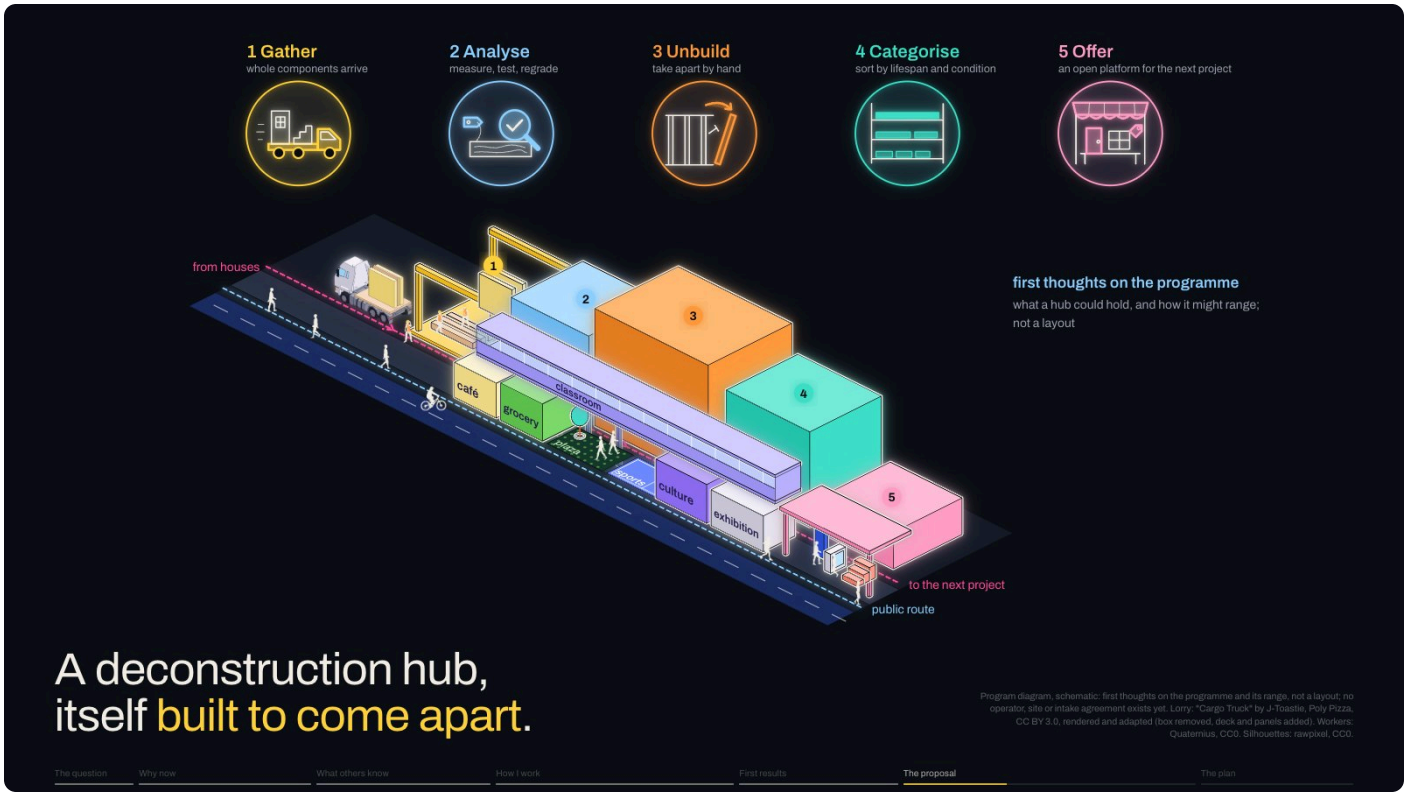


K.118
baubüro in situ, Winterthur
2021, photo Martin Zeller, via ArchDaily

Source note on the slide: Precedent survey, 26 Oct 2026: Habitat for Humanity Greater Vancouver ReStore locations; Lynch 2022 (29 providers, “decentralized and loosely organized”); ReBuilding Center, Portland; Seattle Public Utilities Salvage Lumber Warehouse (opened May 2026); Rotor DC, Brussels; Bauteilbörse Basel; baubüro in situ, K.118, Winterthur 2021. Regrading: Teshnizi 2015, UBC SEEDS. City of Vancouver, Council report, 16 May 2018; Rebuild Hub: Habitat for Humanity Greater Vancouver, 9 Oct 2020, and City of Vancouver, “Reducing construction and demolition waste” (read 26 Oct 2026); ReStore locations, habitatgv.ca (read 26 Oct 2026). Row photos from each organisation; Rotor DC by Pascal Broze; K.118 by Martin Zeller.

26 The deconstruction hub

1. slower So my proposal for GP Two turns the online hub into a building: pause a public deconstruction hub, sketched as first thoughts.
2. Houses arrive in large components, so sites clear in days.
3. Each part is measured and tested, wood regraded, for its next use; keep going
4. unbuilt by hand, indoors; keep going
5. sorted by lifespan and condition; keep going
6. and offered first to the multiplexes replacing them.
7. A multiplex adds homes but no shared room, so the hub could add rooms along the street.
8. A classroom could look into the work, so students and trades learn by watching.

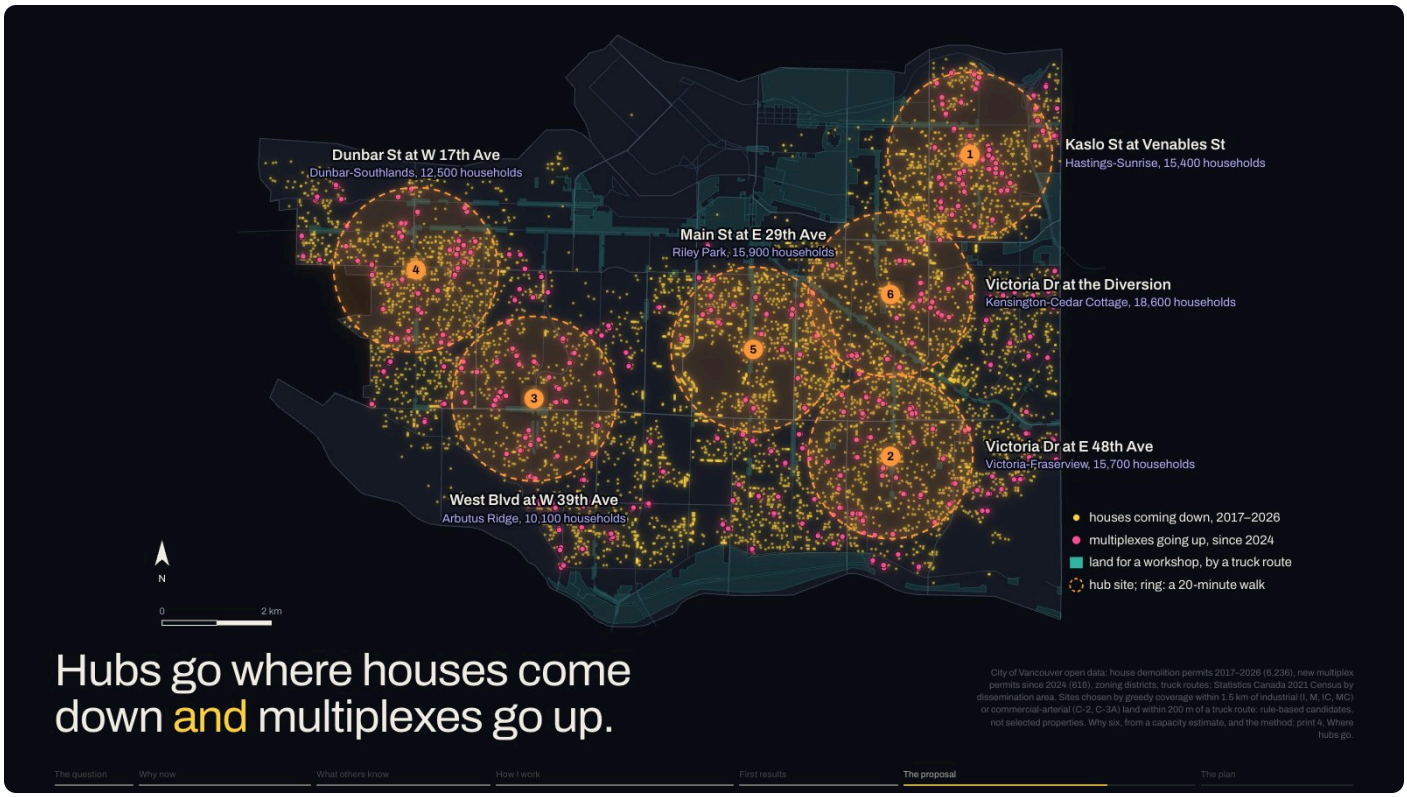


On screen: A deconstruction hub, itself built to come apart.

Source note on the slide: Program diagram, schematic: first thoughts on the programme and its range, not a layout; no operator, site or intake agreement exists yet. Lorry: "Cargo Truck" by J-Toastie, Poly Pizza, CC BY 3.0, rendered and adapted (box removed, deck and panels added). Workers: Quaternius, CC0. Silhouettes: rawpixel, CC0.

27 Where hubs go

1. To place hubs, I mapped three things: keep going
2. where houses come down, keep going
3. where multiplexes go up, keep going
4. and land a workshop could use, by a truck route.
5. Six sites reach most of both within a twenty-minute walk, and over a quarter of city households.

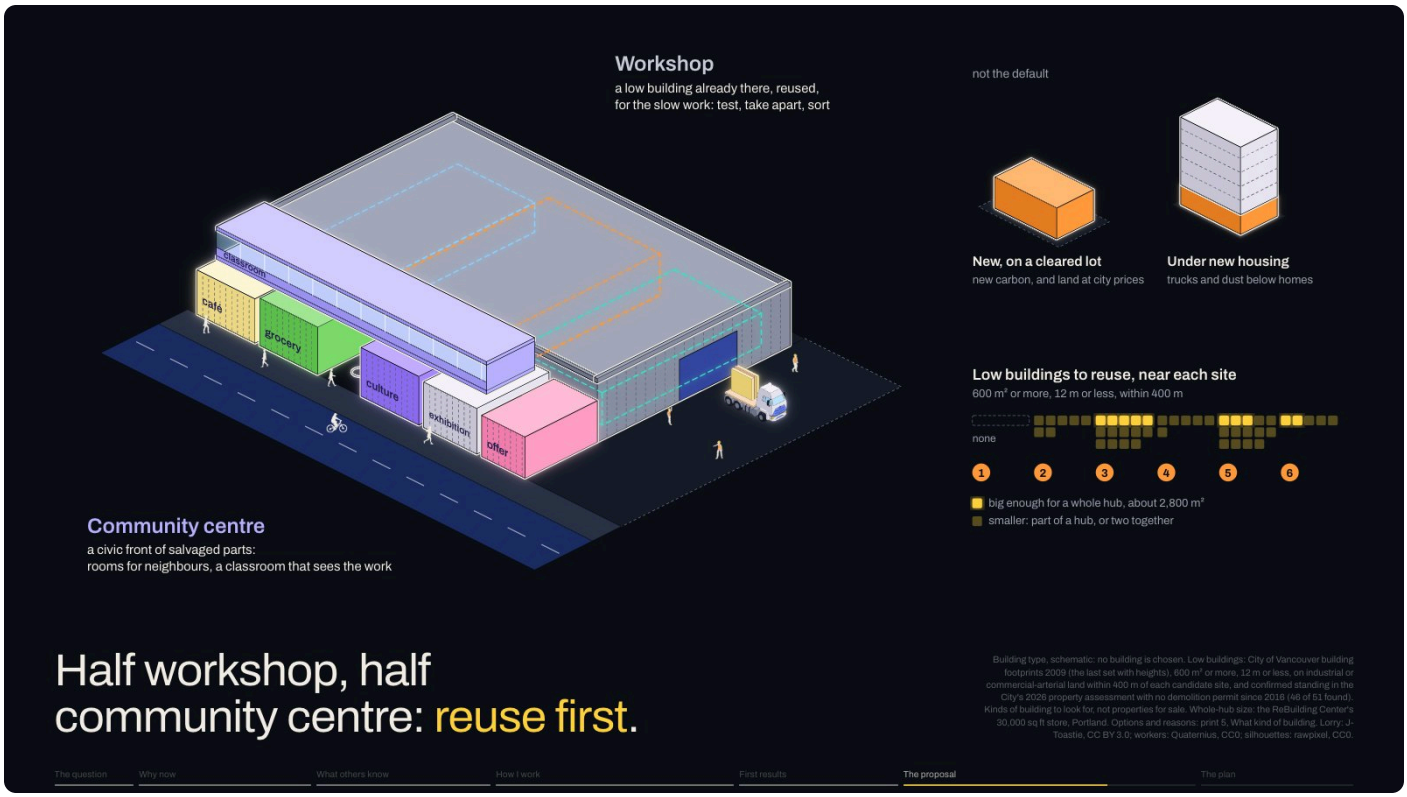


On screen: Hubs go where houses come down and multiplexes go up.

Source note on the slide: City of Vancouver open data: house demolition permits 2017–2026 (6,236), new multiplex permits since 2024 (616), zoning districts, truck routes; Statistics Canada 2021 Census by dissemination area. Sites chosen by greedy coverage within 1.5 km of industrial (I, M, IC, MC) or commercial-arterial (C-2, C-3A) land within 200 m of a truck route: rule-based candidates, not selected properties. Why six, from a capacity estimate, and the method: print 4, Where hubs go.

28 What kind of building

1. What kind of building is it? pause My answer is half workshop, half community centre.
2. Building new, on a cleared lot or under housing, spends carbon and land.
3. So it reuses a low building already there, a store or warehouse, for slow work, keep going
4. and adds a civic front of salvaged parts.
5. Most sites have several such buildings nearby, and this year's City assessments show they still stand.



On screen: Half workshop, half community centre: reuse first.

Source note on the slide: Building type, schematic: no building is chosen. Low buildings: City of Vancouver building footprints 2009 (the last set with heights), 600 m² or more, 12 m or less, on industrial or commercial-arterial land within 400 m of each candidate site, and confirmed standing in the City's 2026 property assessment with no demolition permit since 2016 (46 of 51 found). Kinds of building to look for, not properties for sale. Whole-hub size: the ReBuilding Center's 30,000 sq ft store, Portland. Options and reasons: print 5, What kind of building. Lorry: J-Toastie, CC BY 3.0; workers: Quaternius, CC0; silhouettes: rawpixel, CC0.

29 What it brings to Vancouver

1. It would bring three kinds of resilience.
2. First, time: buildings that change one part at a time.
3. Second, climate: deep eaves, raised bases, replaceable skins keep timber dry.
4. And third, material: parts that can be checked, repaired and moved on, pause which I learned from the pagoda study.

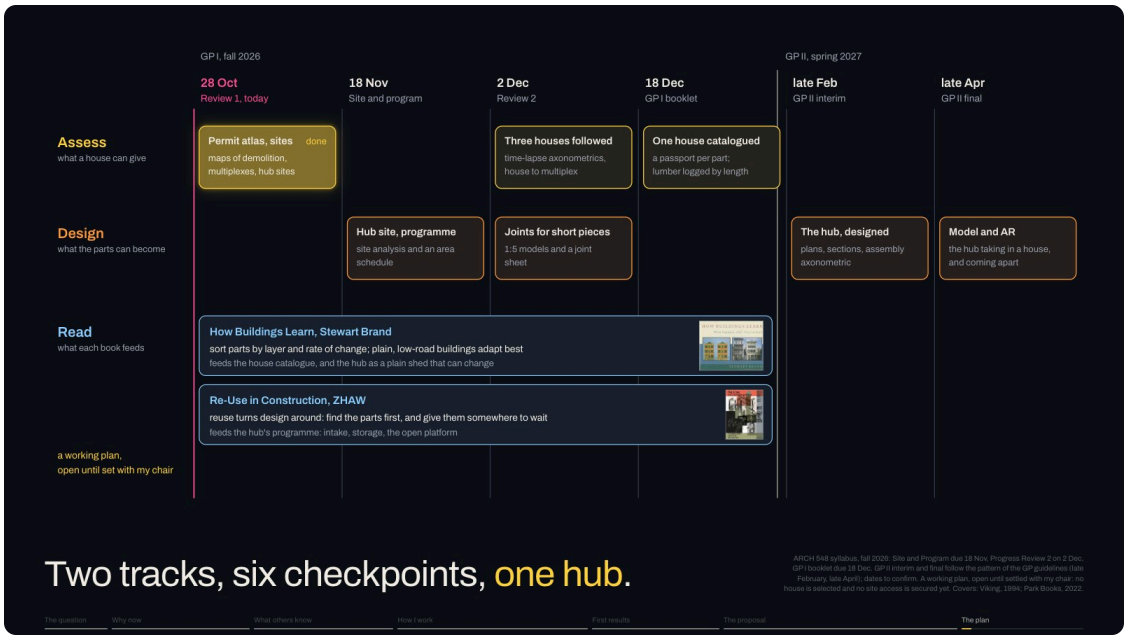


On screen: What it brings: resilience in time, climate and material.

Source note on the slide: Design intentions to test, not results. Timber detailing informed by the pagoda study (Yingxian Pagoda, Shanxi), not copied from it. Drawing: R. Liu with Astra 6.

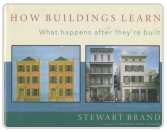
30 The year ahead

1. This plan is still open, pause until I work it through with my chair.
2. To assess, I will follow three houses near the first hub site into their multiplexes, and catalogue one.
3. To design, I will test joints for short lumber, then design the hub.
4. Two readings guide it: Brand, and the Swiss compendium Re-Use in Construction.



On screen: Two tracks, six checkpoints, one hub.

From the sources



How Buildings Learn
Stewart Brand, Viking
1994, cover

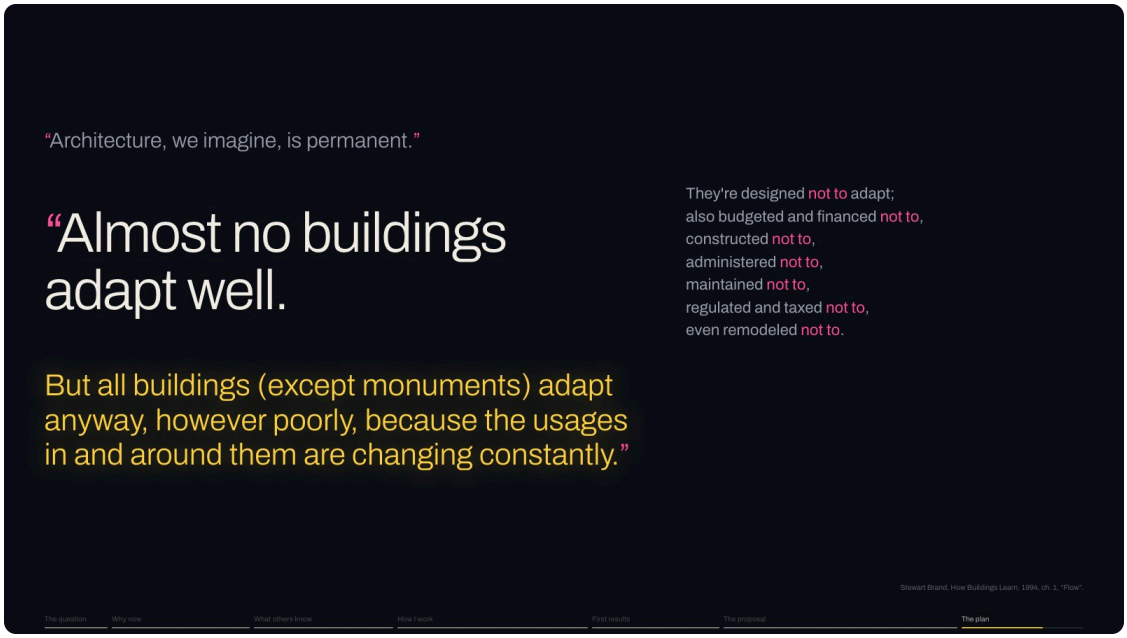


Re-Use in Construction
ZHAW Institute of Constructive Design, Park Books
2022, cover

Source note on the slide: ARCH 548 syllabus, fall 2026: Site and Program due 18 Nov, Progress Review 2 on 2 Dec, GP I booklet due 18 Dec. GP II interim and final follow the pattern of the GP guidelines (late February, late April); dates to confirm. A working plan, open until settled with my chair: no house is selected and no site access is secured yet. Covers: Viking, 1994; Park Books, 2022.

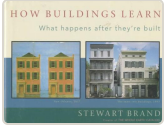
31 Buildings adapt anyway

1. I'll end with Brand: pause “Architecture, we imagine, is permanent.” pause
2. slower “Almost no buildings adapt well.
3. They're designed not to adapt; also budgeted and financed not to, constructed not to, administered not to, maintained not to, regulated and taxed not to, even remodeled not to.
4. But all buildings, except monuments, **adapt anyway**, however poorly, because the usages in and around them are changing constantly.” pause



On screen: Architecture, we imagine, is permanent. Almost no buildings adapt well. They're designed not to adapt; also budgeted and financed not to, constructed not to, administered not to, maintained not to, regulated and taxed not to, even remodeled not to. But all buildings (except monuments) adapt anyway, however poorly, because the usages in and around them are changing constantly.

From the sources

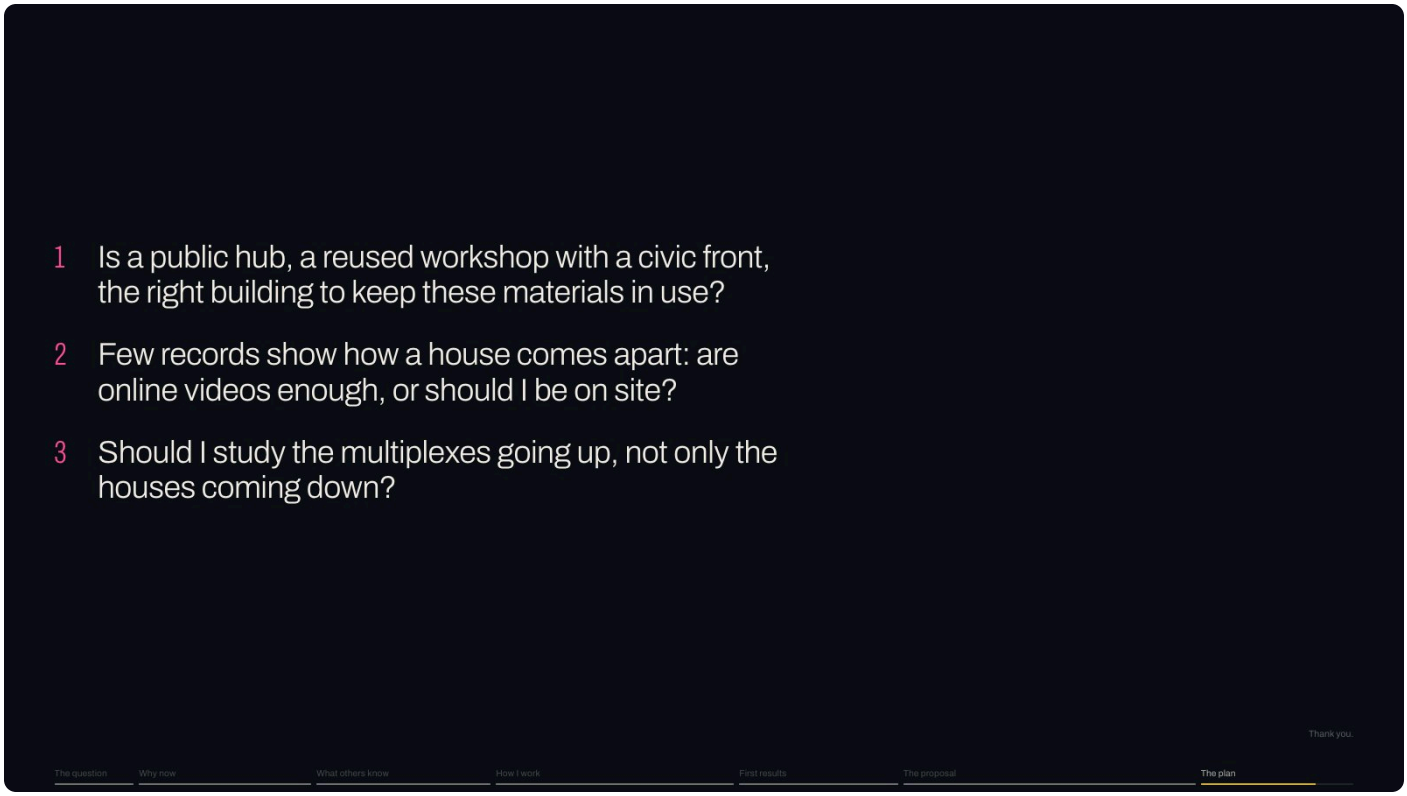


How Buildings Learn
Stewart Brand, Viking
1994, cover

Source note on the slide: Stewart Brand, How Buildings Learn, 1994, ch. 1, “Flow”.

32 Questions

1. slower To close, I have one question, and two doubts of my own.
2. Is this hub, a reused workshop with a civic front, the right building to keep these materials in use?
3. Few records show how houses come apart; are online videos enough, or should I be on site?
4. And should I study multiplexes going up, not only houses coming down? pause Thank you.



On screen:

Source note on the slide: Thank you.